

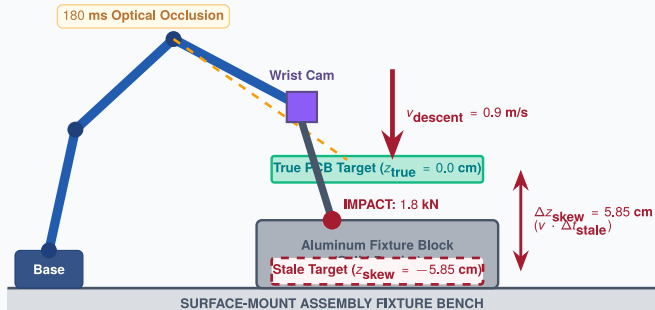
THE STALE FRAME TREE COLLISION: FORENSIC ANATOMY

Uncompensated coordinate transform latency ($\Delta t = 65$ ms) in high-speed surface-mount assembly:
multiplying real-time joint encoders with stale vision transforms injects 5.85 cm spatial skew, destroying ceramic vacuum tooling.

Physical Workcell & Spatial Skew Geometry

Non-Isochronous Transform Multiplication:

$$\mathbf{T}_{\text{target}}^{\text{base}}(t_{\text{now}}) = \mathbf{T}_{\text{cam}}^{\text{base}}(t_{\text{now}}) \cdot \mathbf{T}_{\text{target}}^{\text{cam}}(t_{\text{now}} - 65 \text{ ms})$$



Four-Stage Flight-Recorder Timeline

$T - 0.245$ s **NOMINAL ACQUISITION (30 Hz)**

Wrist RGB-D camera captures valid PCB alignment fiducials ($t_{\text{exp}} = 10$ ms). Transform $\mathbf{T}_{\text{target}}^{\text{cam}}$ is valid and latched in camera frame.

$T - 0.180$ s **FOREARM LINK OCCLUSION**

Arm linkage rotates into optical path, occluding camera for 180 ms ($\phi_t = \emptyset$). Transform buffer stops updating; no active TTL lease check exists.

$T - 0.065$ s **STALE TRANSFORM MULTIPLICATION**

State engine multiplies real-time joint encoders (t_{now}) by cached 65 ms old transform. Injects vertical skew: $\Delta z = 0.9 \text{ m/s} \times 0.065 \text{ s} = 5.85 \text{ cm}$.

$T + 0.000$ s **1.8 kN MECHANICAL CRASH**

Trajectory planner descends targeting phantom coordinates. Vacuum tooling shatters against aluminum block (1.8 kN dynamic impact, \$3,200 damage).

ARCHITECTURAL REMEDIATION: PTP-INDEXED RING BUFFERS & EXPIRING STATE LEASES (STATE-01)

Static SRAM Ring Buffers ($\text{malloc} = 0$)

Transforms stored in fixed circular buffers indexed by 64-bit IEEE 1588 PTP hardware timestamps.

$\mathfrak{s} \in \mathfrak{c}(3)$ Geodesic Interpolation

State engine evaluates query poses via exact geodesic interpolation in Lie algebra tangent space.

Hard State TTL Leases ($\tau_{\text{TTL}} = 100$ ms)

If transform age exceeds τ_{TTL} , engine emits `TRANSFORM_EXPIRED` triggering safe stop.